

Function / Concept	Purpose	Usage in Coffee Shop Sales Project
STR_TO_DATE()	Converts text into a valid date/time format.	Used to convert imported transaction dates and times from text format into MySQL date/time format for accurate analysis.
SUM()	Adds values in a column.	Used to calculate total sales revenue and total quantity sold.
COUNT()	Counts the number of records.	Used to determine total orders and transaction counts.
AVG()	Calculates the average value.	Used to find average daily sales and benchmark performance.
ROUND()	Rounds numbers to a specified number of decimal places.	Used to format KPI values, percentages, and sales figures for reporting.
MAX()	Returns the highest value.	Used to identify the highest sales day, month, or product performance.
MIN()	Returns the lowest value.	Used to identify the lowest sales day or underperforming periods.
MONTH()	Extracts the month from a date.	Used for monthly sales, orders, and quantity trend analysis.
DAY()	Extracts the day from a date.	Used to analyze daily sales patterns and create calendar-based reports.
DAYOFWEEK()	Returns the weekday number.	Used to compare weekday and weekend sales performance.
HOUR()	Extracts the hour from a time value.	Used to identify peak business hours and customer purchasing trends.
LAG()	Retrieves values from previous rows.	Used for Month-over-Month (MoM) comparison of sales, orders, and quantity sold.
CASE Statement	Applies conditional logic (IF-ELSE).	Used to classify days as Above Average, Below Average, or Average sales days.
GROUP BY	Groups records based on a column.	Used to summarize sales by month, day, store location, product category, and weekday.
ORDER BY	Sorts results in ascending or descending order.	Used to rank products, sort dates, and display trends chronologically.
JOINS	Combines data from multiple tables.	Used when integrating sales data with product, store, or date-related information.
SUBQUERIES	A query nested inside another query.	Used to compare sales against averages and perform intermediate calculations.